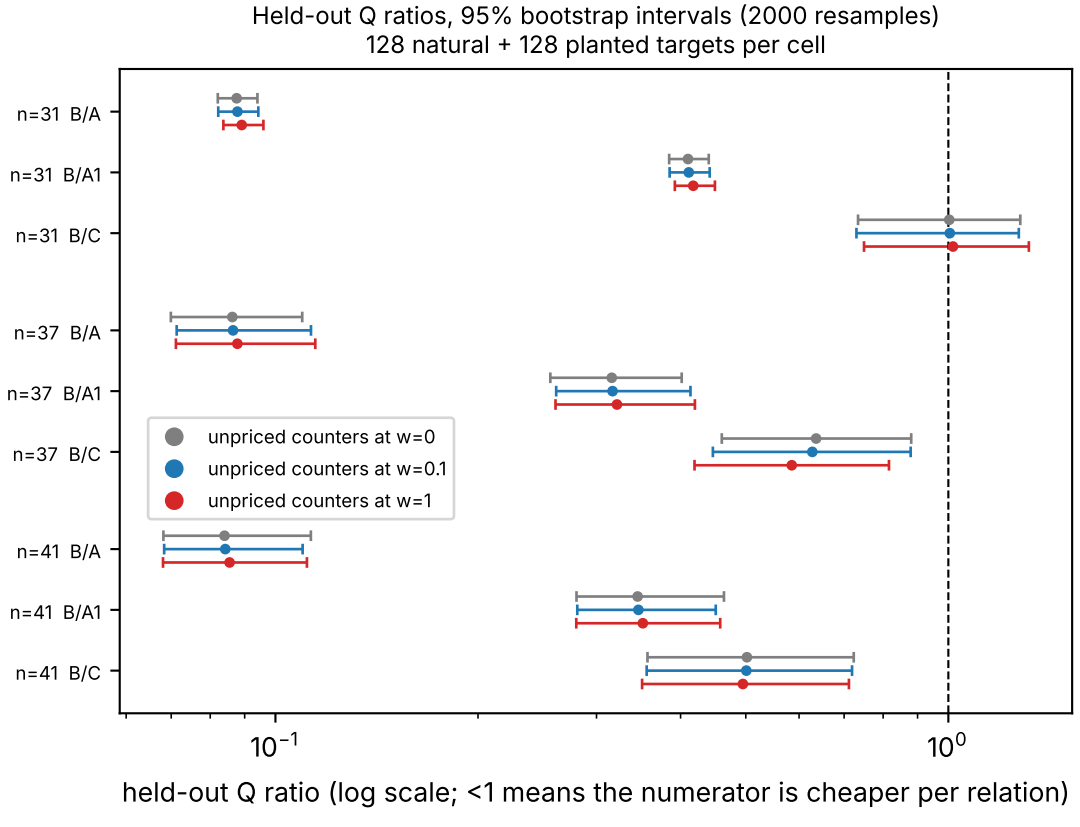
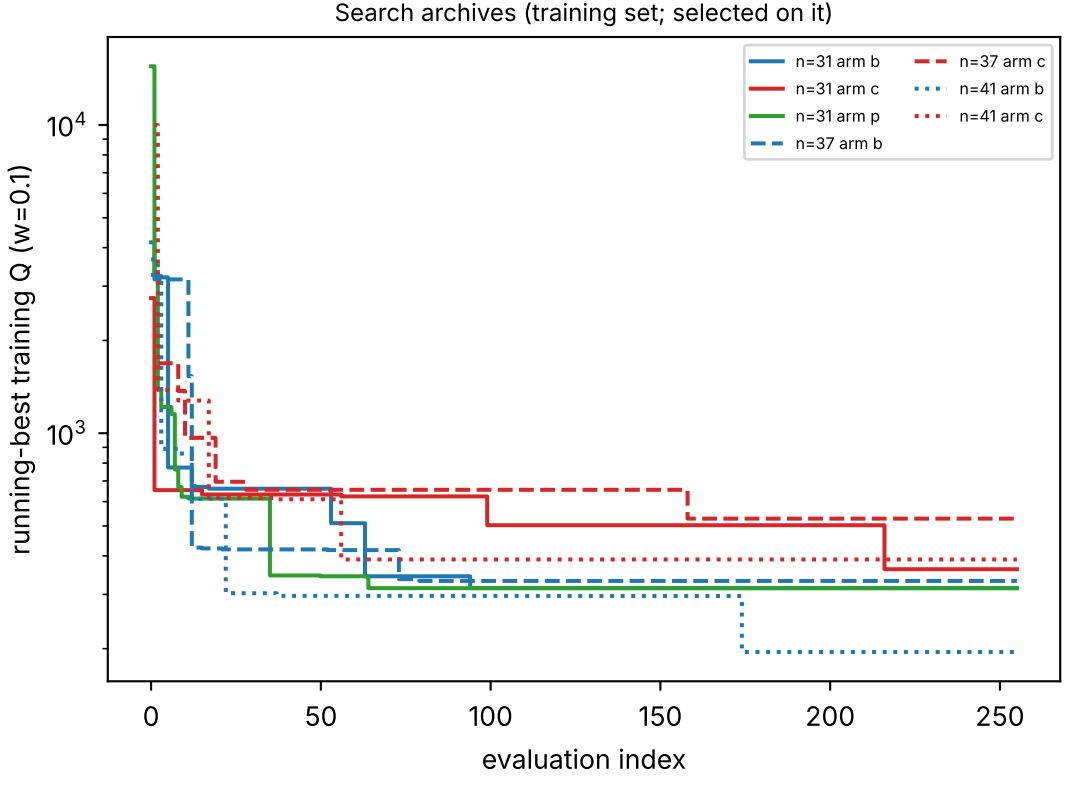
Item 10 kernel program search, held-out assessment: per-cell verdicts n=31 MATCHES, n=37 ADVANCES, n=41 ADVANCES; overall INCONCLUSIVE (cells disagree)





Item 10: program search over the decomposition kernel (final assessment, after repair pass R2 stage 3)

Verdict: INCONCLUSIVE (level: constant), with per-cell verdicts MATCHES (n=31), ADVANCES (n=37), ADVANCES (n=41)

The frozen rule (`PROTOCOL.md`, "Decision rule") was applied to the held-out data (128 natural + 128 planted targets per cell, 2000-resample bootstrap intervals) by the instrument's own `summarize` command (`raw-r2/summary.txt`, `raw-r2/summary.json`). It gives one verdict per cell, and the overall verdict is the common one only if the three cells agree. They do not, so the rule's overall verdict is **INCONCLUSIVE**. No cell regresses or trades, and no control failed.

What is measured, stated plainly (replica of the S_4 pair-and-root kernel, toy bases, l = 9, GF(2^n), n = 31, 37, 41; counted field multiplications and squarings plus the four unpriced native counters priced at w = 0, 0.1, 1; quantity Q = counted cost per relation times the number of columns, held-out):

* The search finalist B has a lower Q than the incumbent A in all three cells and at all three weights, with the whole 95% interval below 1: Q(B)/Q(A) at w = 0.1 is **0.088 [0.082, 0.094]** (n=31), **0.086 [0.071, 0.113]** (n=37), **0.084 [0.068, 0.110]** (n=41). That is about 11 to 12 times fewer counted operations per relation in this replica. It is a measured improvement over the incumbent baseline of the program space at constant-factor level, and it is also what the rule calls "better than A".
* Whether the *search* produced it is the other half of the rule. Against random search of the same 256 evaluations, Q(B)/Q(C) at w = 0.1 is 1.005 [0.730, 1.273] at n=31 (straddles 1: MATCHES), 0.628 [0.447, 0.879] at n=37 and 0.501 [0.356, 0.719] at n=41 (better at all weights: ADVANCES). One cell in three is a tie.
* A large part of the gain does not need the search. The added one-gene reference A1 (incumbent plus `lift = 1, stop = 1`: do not scan abscissae that have no point) alone gives Q(A1)/Q(A) = 0.213, 0.273, 0.243. What B adds beyond that patch is Q(B)/Q(A1) at w = 0.1 = **0.411 [0.385, 0.442]**, **0.317 [0.261, 0.414]**, **0.346 [0.281, 0.451]** (all better). The next section says where that extra factor comes from.
* Level: **constant**. No program in this space changes the Theta(2^n) relation-collection cost.

Distance to matched rho: **unchanged, and not measured by this item.** It is a one-stage diagnostic; it computes no S, no rho ratio and no GAE, so there is no cross-method row to rank under AGENTS.md section 5a. The kernel's unpriced native counters (word XORs, hash probes, trace tests, pair iterations) are priced at 0, 0.1 and 1 counted operations inside Q for every arm, the same for every arm. The ratios do not move with that price (Q(B)/Q(A): 0.0876, 0.0878, 0.0891 at w = 0, 0.1, 1 at n=31; the other cells differ by under 0.002), so no conclusion depends on the price. A constant-factor speedup of the relation phase leaves the exponent, and with it the gap to Pollard rho at any cryptographic size, as it was.

Per-cell result (held-out, from `raw-r2/summary.txt`; each cell is "better", "worse" or "straddles 1" at all of w = 0, 0.1, 1)

cell	B (search finalist, genes)	Q(B)/Q(A), w=0.1 [95%]	Q(B)/Q(C), w=0.1 [95%]	Q(B)/Q(A1), w=0.1 [95%]	B vs A	B vs C	cell verdict
n=31	affine-aligned-s2, lift1 stop0 root horner, q=b1, sqtab, abort 1/2, batch 8 rows (idx 94; same program as the p arm's idx 64)	0.088 [0.082, 0.094]	1.005 [0.730, 1.273]	0.411 [0.385, 0.442]	better at all 3	straddles at all 3	MATCHES
n=37	affine-aligned-s1, lift1 stop1, horner, b1, sqtab, abort 1/8, batch 8 rows (idx 79)	0.086 [0.071, 0.113]	0.628 [0.447, 0.879]	0.317 [0.261, 0.414]	better at all 3	better at all 3	ADVANCES
n=41	the same program as n=37 (idx 174)	0.084 [0.068, 0.110]	0.501 [0.356, 0.719]	0.346 [0.281, 0.451]	better at all 3	better at all 3	ADVANCES

Held-out Q at w = 0 / 0.1 / 1 (measured), incumbent A, A1, B, C:

cell	A	A1	B	C
n=31	3910 / 4132 / 6126	834.5 / 881.8 / 1307	342.4 / 362.8 / 545.8	341.3 / 360.9 / 537.2
n=37	3089 / 3264 / 4839	843.1 / 890.9 / 1320	266.5 / 282.3 / 424.8	419.0 / 449.7 / 726.0
n=41	3467 / 3663 / 5431	843.4 / 891.1 / 1321	291.2 / 308.5 / 464.2	580.2 / 616.0 / 937.7

Training-to-held-out shrinkage of the finalists' Q at w = 0.1 (held-out / training, measured): B 1.15 (n=31), 0.85 (n=37), 1.58 (n=41); C 1.00, 0.85, 1.58. The n=41 finalists pay a 58% winner's-curse penalty (6 natural training targets); the rule does not use training values.

Where the gain comes from (derived from `raw-r2/heldout-n*.jsonl`, `raw-r2/assessment-tables.json`)

Q = cbar_N . B_r / (T_eff . phi). Held-out factors for Q(B)/Q(A) at w = 0.1, product equal to the reported ratio
to four digits:

cell	counted cost per natural target B/A	B_r B/A	1/T_eff factor	1/phi factor
n=31	0.185	1.093	0.384	1.133
n=37	0.064	1.000	0.500	2.723
n=41	0.067	1.091	0.385	2.977

* The base family: all six finalists (b and c, three cells) are `affine-aligned` bases (`U` inside ker Tr, `delta`
  identically 0). Their credit is T_eff (the number of trace-feasible triples), 2.6, 2.0 and 2.6 times the
  incumbent's. That is the trace-class lever the protocol already named, via the yield model, not a cheaper kernel.
  Its price is retention of planted relations: phi falls to 0.88, 0.37, 0.34.
* The counted cost per natural target falls by 5.4x (n=31), 15.6x (n=37) and 14.9x (n=41) against A, of which the
  lift prefilter alone (A1) is 4.7x, 3.7x and 4.1x. The abort gene (1/2 at n=31, 1/8 at n=37, 41) cuts the pairs scanned and
  gives up planted relations: reverting it at n=37 raises held-out phi from 0.375 to 1.0 and Q by 1.58 times, so
  inside Q the trade is a net gain, but only because Q divides by phi; a pipeline that needs every relation
  scanned would not take this trade. The quartic, square-table and batching genes change Q by 1% or less in each cell.
* Against A1 the per-target cost of B is 0.865 (n=31), 0.233 (n=37), 0.277 (n=41), so most of B's extra factor
  over A1 at n=31 is the trace-class credit, and at n=37, 41 it is the abort trade plus the credit.
* Ablation (held-out 48 + 48 targets, w = 0.1, attribution only, no verdict; `raw-r2/ablate-n*.jsonl`): reverting
  the family to `sub-low` doubles B's Q (2.05x, 2.01x, 2.01x); reverting `lift` multiplies it by 3.9x, 4.1x, 2.9x;
  reverting `abort` by 1.19x, 1.58x, 1.40x; every other gene is within 1%. The incumbent with only one gene
  of B changed has Q/Q(A) = 0.21, 0.27, 0.24 for `lift`; 0.42, 0.50, 0.42 for `fam`; 0.84, 0.59, 0.75 for `abort`;
  and 0.987 or more for `quartic`, `sqtab`, `batch` and `stop`. The random-search finalists C show the same pattern with `root = eval` and `pairs = full` as the
  costly genes it carries (reverting `root` lowers C's Q by 0.64x at n=37, which is the search's advantage there).

Method (unchanged from PROTOCOL.md)

(1+1)-ES with elite archive (b), 256 distinct random valid programs (c, negative control by design), the same engine from a handicapped start (p, n=31 only), 256 evaluations each, on 6 natural + 32 planted training targets;
finalist = lowest finite training Q. Held-out: 128 natural + 128 planted targets per base, different seeds, no cap.
Bootstrap: 2000 resamples, natural targets resampled jointly across programs, planted targets independently.
Instrument `examples/blindspots_10_kernel_search.rs`, sha256 `f69bbce49c64...` (checked now, unchanged since the first run). Every card builds its own binary from that source and removes it.

Controls

* Instrument correctness (first run, `raw/selftest.txt`, kept): 400 random programs against brute force at n=17,
  l=6, bad = 0; S_4 vanishes on x(P1 +- P2 +- P3) at n = 17, 31, 41; handicap checks all PASS.
* Positive control for the search (arm p, n=31): **PASS** on training (best 314.3 against incumbent 4162.2 and
  half of its own start 15505) and held-out (P is the same program as B: Q(P)/Q(A) = 0.088 [0.082, 0.094]).
  Note p reached the n=31 finalist at idx 64, b at idx 94, so two independent starts converge on one program.
* Negative control: arm c is complete; its held-out ratios against A are better in all three cells (Q(C)/Q(A) 0.087, 0.138, 0.168 at
  w = 0.1), so random search over this space also finds most of the gain; `affine-anti` never has a finite
  training Q in any archive and has zero found relations in both conservation cells (found 0/4000, 0/3000).
* Null: the incumbent on two held-out halves covers 1 at all weights in all cells (n=31 0.985 [0.956, 1.000];
  n=37, 41 1.000 [1.000, 1.000]). The n=37 and n=41 intervals are degenerate because the incumbent scans the
  same pairs on every natural target without a hit, so its cost per target is constant; the check cannot fail
  there, it is weak evidence about noise at those cells and is stated as such.
* Yield conservation and yield model (cards 15, 21; `raw-r2/conserve-17-6.jsonl`, `conserve-19-7.jsonl`): at (17, 6)
  the median kappa over the sub-* families is 1.052e-4 and the finalists' families are within 25% of it
  (affine-aligned-s1 1.040e-4, s2 1.158e-4 (+10.0%)). At (19, 7) kappa lies between 2.74e-5 and 3.06e-5 over the 17
  exact programs run, the two affine-aligned families at 3.06e-5 and 2.85e-5. The model holds, so no finalist is
  disqualified on yield. The non-exact finalist programs (with abort) show lower kappa (3.8e-5 at (17, 6)); that is
  retention loss, not a model violation, and it is what phi measures.
* Dense-cell cost per relation `W` at w = 0.1 (measured, natural yield, no planting, no verdict weight): at
  (17, 6) A 2.300e7, B-n31 2.230e6 (0.097 of A), B-n37 = B-n41 1.991e6 (0.087), C-n31 2.138e6, C-n37 3.074e6,
  C-n41 3.484e6; at (19, 7) A 1.179e8, B-n31 9.864e6 (0.084), B-n37 = B-n41 8.594e6 (0.073), C-n31 9.113e6, C-n37
  1.337e7, C-n41 1.520e7. The dense cells agree in direction and size with the sparse held-out Q.

Unpriced native counters (held-out natural-target means; `raw-r2/assessment-tables.json`)

Per natural target: A scans 1.30e5 to 1.31e5 pairs with 1.03e7 to 1.04e7 word XORs (core 1.84e7 to 1.86e7); A1 2.8e4 to 3.6e4 pairs; B 2.5e4, 8.5e3, 9.0e3 pairs; C 9.6e3, 8.5e3, 9.6e3. Hash probes and trace tests are 0 to 0.1 per target for every arm (witnesses are rare), so the three prices change `lin` and `pairs` only, `lin`
being about 0.55 to 0.75 of `core` in every arm. Hence the weight-independence of the ratios above.

Deviations and process notes

* **No PROTOCOL deviation.** Instrument unchanged, no bug proved; the only change over the first run is the card
  layout (`PROTOCOL.md`, "R2 stage 3").
* **Card 22-r2 (summarize) was not frozen or run through a card.** It binds files that stage 3 produced, which are
  committed now. I ran the instrument's own `summarize` directly (same arguments as `make_r2_cards.py` stage 4:
  `--dir raw-r2 --ns 31,37,41 --resamples 2000 --out raw-r2/summary.json --state raw-r2/state`) from a binary built
  from the sha256-unchanged source. It is a deterministic function of committed files; the outputs are
  `raw-r2/summary.txt` (sha256 `73c58c81cbdc...`) and `raw-r2/summary.json` (`e34f26149...`). No receipt
  exists for it. The derived tables (`assessment-tables.json`, the decomposition and ablation sections) are
  computed by me in Python from the committed JSONL and are labelled derived; they carry no verdict weight.
* **Relays.** Saved as `raw/relay-14.txt` (card 14-r2), `relay-15` (15-r2), `relay-16` (21-r2), `relay-17` (16-r2),
  `relay-18` (17-r2), `relay-19` (18-r2), numbered after the earlier stages' relay-1..13, which are not overwritten.
  `check_relay` on all six: `relay_verbatim: true`, no findings, status completed. The tool's command-line
  `check-relay` refuses first because its card lint now fails on all six cards ("names repository file ... that binds
  does not pin": each card names the output file it created, which exists now and is not bound). This is the
  same after-the-fact lint failure card 05-r2 showed, a property of the card text, not of the run; the cards

were
  linted CARD OK at freeze time and every receipt is completed, so I called `check_relay` directly. Two of my
  own transcriptions (relay-14, relay-19) wrote a `rmdir` argument with a stray `/b10`; corrected against the
returned
  text before checking, not an executor deviation.
* **Execution order.** The cards ran one after another; 14, 15 and 21 wrote `finalists-n*.json` with identical
bytes
  (sha256 `9f86775c...`, `58525a25...`, `6092cbe3...`, equal in every relay).
* **Host L0.** Shared 4-vCPU container, up to two lanes at once; wall times are descriptive only and never
enter Q.

Caveats

* Toy scale: GF(2^n), n = 31..41, l = 9, sparse regime (about 0.02 raw S_4 roots per natural target); the
yield and
  the cost per relation of natural targets are carried to these cells by the yield model checked at (17, 6)
and
  (19, 7). A replica of the incumbent kernel counting multiplications unfused, not the crate's
`SubspaceOracle`
  (Deviation 1); absolute counts overstate real field work. Nothing here transfers to cryptographic sizes, and
no
  GAE is attempted.
* Most of the measured gain is `lift` (4x), the trace-class base choice (2 to 2.6x, a lever the protocol named
  beforehand) and an early-abort trade that gives up relations; the search found them, and so did random
search at
  n=31 and, partly, n=37 and 41. The part the search adds over random is 0.63 and 0.50 of C's Q at n=37 and
41, and
  nothing at n=31. Three cells, one program shared by two of them, so the evidence for the search over random
is
  weaker than the three-cell count suggests.
* The n=41 finalists' Q rose 58% from training to held-out; the n=31 B finalist's natural yield (4/128 held-
out)
  is above the 2% under which `Q`'s first-order error is stated to stay, so its error term is not negligible
there.
* This is not an end-to-end relation-phase result: no real factor-base size, no linear algebra, no final DLP.
* Relation to the owner's success criterion: B improves on the incumbent baseline by a verified measured
amount in
  the counted unit of this replica, in every cell; that is reported as a constant-factor program-space
improvement,
  not as an improvement of the exponent and not as progress against rho.

Files

`graph.svg`, `graph.png` (source `make_graph_final.py`, data `raw-r2/summary.json` and
`raw-r2/state/*.jsonl`),
`REPORT.pdf`, `raw-r2/summary.txt`, `raw-r2/summary.json`, `raw-r2/assessment-tables.json`,
`raw-r2/heldout-n*.jsonl`, `raw-r2/ablate-n*.jsonl`, `raw-r2/conserve-*.jsonl`, `raw-r2/finalists-n*.json`,
`raw-r2/receipt-card-*-r2.json`, `raw/relay-1..19.txt`. A copy of the PDF, SVG and PNG is in
`benchmarks/20261018-blindspots-kernel-program-search/` (the `benchmarks/README.md` row is the integration
step's).

First run (superseded)

The assessment below is the previous one (R2 stage 2: all seven searches complete, no held-out data), kept
verbatim. Its verdict was INCONCLUSIVE because the held-out, ablation and conservation cards had not run. It
is
superseded by the verdict above.

Item 10: program search over the decomposition kernel (assessment after repair pass R2, stage 2)

Verdict: INCONCLUSIVE (level: constant; the frozen rule still has no input)

The frozen decision rule (`PROTOCOL.md`, "Decision rule") is defined on **held-out** data (128 natural + 128 planted targets per cell, 2000-resample bootstrap intervals for `Q(B)/Q(A)` and `Q(B)/Q(C)` at w = 0, 0.1, 1), and on the controls (null, yield conservation, a finalist's family passing the yield model). The held-out, finalist, ablation, conservation and summary cards (14-18, 21-r2, 22-r2) have not been frozen or run, so no per-cell ADVANCES / MATCHES / REGRESSES / TRADES can be assigned; the rule's own "otherwise" branch applies. This
assessment adds what stage 2 delivered: **all seven searches (b and c at n = 31, 37, 41; p at n = 31) are now complete at 256 evaluations, every receipt completed, every relay verbatim, every archive hash equal to the receipt's output hash.** Nothing here is a speedup claim and nothing is credited to the search.

What changed against the previous assessment (below): the n41 arm-b search was **not** at a plateau at 128; its
best training Q fell from 296.4 to 195.1 (evaluation 174). The earlier sentence "all three search arms reach their
plateau by evaluation 128" is withdrawn.

Distance to matched rho: **unchanged and not measured by this item.** It is a one-stage diagnostic (S_4 pair-and-root kernel, toy factor bases, l = 9, n = 31, 37, 41); it computes no S, no rho ratio and no GAE, so the
section 5a band (0, 0.1, 1 GAE per unpriced native operation) has no cross-method row to rank. The kernel's unpriced native counters (word XORs, hash probes, trace tests, pair iterations) are priced at 0, 0.1 and 1 counted
operations inside Q for every arm, and the training ratios below hold at all three weights (they differ by under
one part in ten between w = 0 and w = 1). Even a full success would be a constant factor on the relation phase,
not the exponent; the distance to rho is whatever it was before this item.

Method (unchanged from PROTOCOL.md)

Counted fitness Q(pi, w) = cbar_N(w) . B_r / (T_eff . phi) on 6 natural + 32 planted training targets; (1+1)-ES with
elite archive (b), 256 distinct random valid programs (c, negative control by design), the same engine from a handicapped start (p, n = 31). Incumbent = evaluation 0 of the arm-b archive. Stage 2 resumed each chain from its
committed 128-line archive (bound by sha256, copied with `cp -t`) and appended 128 more evaluations; the first 128 lines of every new archive are byte-identical to the stage-1 archive (checked), as the first 192 lines of n31-b are to the card-04 archive.

Numbers (TRAINING set only; measured, from `raw-r2/state/*.jsonl`)

"ratio" = best finite training Q of the arm / the incumbent's Q on the same targets. These are selected-on quantities (best of 256 on 6 natural targets), carry no interval, and are exactly what the rule forbids reading as
a verdict.

cell, arm	evals	best program (training)	idx	Q at w=0.1	ratio at w = 0 / 0.1 / 1	phi	Q=inf (of which dominated)
n31 b	256	affine-aligned-s2 lift1 stop0 root0 q=b1 sqt1 ab=2 bt=1	94	314.3	0.0753 / 0.0755 / 0.0767	0.94	10 (2)
n37 b	256	affine-aligned-s1 lift1 stop1 root0 q=b1 sqt1 ab4 bt1	79	331.8	0.1014 / 0.1016 / 0.1031	0.31	10 (5)
n41 b	256	affine-aligned-s1 lift1 stop1 root0 q=b1 sqt1 ab4 bt1	174	195.1	0.0531 / 0.0533 / 0.0541	0.53	2 (1)
n31 c	256	affine-aligned-s1 lift1 stop0 root0 q=general sqt0 pm=full ab4 bt0	216	362.1	0.0869 / 0.0870 / 0.0873	0.31	72 (53)
n37 c	256	affine-aligned-s1 lift1 stop0 root1 q=general sqt1 ab4 bt1	158	528.4	0.1594 / 0.1619 / 0.1763	0.31	89 (71)
n41 c	256	affine-aligned-s1 lift1 stop0 root1 q=b1 sqt1 pm=full ab4 bt2	56	389.5	0.1058 / 0.1063 / 0.1092	0.53	82 (72)
n31 p	256	affine-aligned-s2 (same program as n31 b idx 94, found at idx 64)	64	314.3	start 15505 -> 314.3	0.94	5 (2)

Incumbent training Q at w = 0.1: n31 4162.2, n37 3264.2, n41 3663.4 (phi = 1).
Search versus random at equal evaluations (256 each, training, w = 0.1), best-b / best-c: **0.868 (n31), 0.628 (n37), 0.501 (n41)**. Descriptive, no interval, same targets used for selection in both arms; it is the quantity
that the held-out `Q(B)/Q(C)` comparison will test and it does not yet support ADVANCES over MATCHES. At n31 the
two arms are within 14% of each other on training data and the gap could be selection noise.
Search consumes far fewer capped evaluations than random (Q=inf: 2-10 versus 72-89 of 256), which is a cost-of-
search observation, not a quality-of-finalist one.

Running-best training Q (w = 0.1): n31 b 4162 -> 314 (by 128, unchanged to 256); n37 b 3264 -> 332 (by 128, unchanged); n41 b 3663 -> 296 (128) -> 195 (256); c: n31 503 -> 362, n37 655 -> 528, n41 389.5 (unchanged).

Controls (state as of now)

* Instrument correctness (selftest, `raw/selftest.txt`, first run, kept): 400 random programs against brute force at
  n = 17, l = 6, bad = 0; S_4 vanishes on x(P1 +- P2 +- P3) at n = 17, 31, 41; handicap checks all PASS.
* **Positive control for the search (arm p), now final at 256 evaluations: PASSES on training data.** Best training Q
  314.3 is at most the incumbent's 4162.2 and at most half of its own start Q 15505 (ratio 0.0203).
* Negative control, arm c: complete. `affine-anti` never has a finite Q in any archive (every occurrence is
  `inf`), as predicted. Random search over this program space already finds large training gains (0.087, 0.162,
  0.106 of the incumbent), so most of the training gain is the space, not the search.
* Not run: null (incumbent on two held-out halves), yield conservation at (17,6) and (19,7), finalists, ablation,
  reference arm a1. A finalist whose family violates the yield model cannot support a verdict; that check has not
  happened.

Observations that the held-out stage must test (not findings)

1. All six arm-b and arm-c winners at l = 9 are `affine-aligned` bases with `lift = 1`. The `T_eff` credit this gives
   in Q (n31: 2.90e6 versus 1.11e6; n37: 3.21e6 versus 1.60e6; n41: 3.50e6 versus 1.35e6; factors 2.6, 2.0, 2.6)
   is the yield model, not a cheaper kernel. Whether the model holds at those bases is the conservation check.
2. `lift = 1` (do not scan abscissae without a point) is shared by every winner; how much of the gain it alone
   explains is the a1 reference arm (card 14), not yet run.
3. The aligned winners give up planted retention (phi 0.94, 0.31, 0.53 for the b winners): Q divides by phi, and
   held-out phi is what counts. The n41 winner changed between 128 and 256 evaluations, so the finalist is not
   stable under search length at that cell.

Deviations and caveats

* Staging, not a protocol change: outputs under `raw-r2/`, searches split in two 128-evaluation cards
  (`PROTOCOL.md` "R2 stage 2"); question, arms, cells, metric and rule unchanged.
* Relay files for this stage are `raw/relay-8..13.txt` (relay-1..7 belong to earlier stages and are not
  overwritten). `check-relay` on all six: `relay_verbatim: true`, no findings. Two of my first transcriptions
  (relay-9, relay-13) failed with one missing line each; both were my own copying errors (a gene digit and a
  dropped trailing line), corrected against the returned text and then verified; they are not executor deviations.
* Card 05-r2's after-the-fact lint failure (see below) was fixed in the stage-2 card text; the stage-2 cards pass.
* Host L0 (shared 4-vCPU container, two lanes at once): wall times (87-291 s per 128) are descriptive only.
Toy scale:
  GF(2^n), n = 31..41, l = 9, sparse regime (about 0.02 raw S_4 roots per natural target); a replica of the
  incumbent kernel, not the crate's `SubspaceOracle` (Deviation 1); no result transfers to cryptographic sizes.
* Not a performance result in the sense of AGENTS.md: no held-out measurement exists, so nothing was copied to
  `benchmarks/` and the leaderboard was not regenerated.

Next steps (not done here)

Freeze stage 3 (`make_r2_cards.py 3`: cards 14-18 and 21-r2, which bind the seven archives above by sha256; they
can be frozen now), run and commit them, freeze stage 4 (card 22-r2), then apply the rule to the held-out summary
and replace this verdict.

Files

`graph.svg`, `graph.png` (source `make_graph.py`, data `raw-r2/training-stage-summary.json`), `REPORT.pdf`,
`raw/relay-1..13.txt`, `raw-r2/receipt-card-*-r2.json`, `raw-r2/state/*.jsonl`, `raw-r2/state-1/*.jsonl`.

First run (superseded)

Previous assessment (R2 stage 1, kept verbatim below; its verdict was INCONCLUSIVE and so is this one, but its
numbers for n37/n41 arm b and all arm c are superseded by the 256-evaluation table above, and its plateau claim
is withdrawn):

>
Verdict: INCONCLUSIVE (level: constant; no verdict can be computed yet)

The frozen decision rule (`PROTOCOL.md`, "Decision rule") is defined on **held-out** data
(128 natural + 128 planted targets per cell, 2000-resample bootstrap intervals for
`Q(B)/Q(A)` and `Q(B)/Q(C)` at w = 0, 0.1, 1). The held-out, finalist, ablation, conservation and
summary cards (14-18, 21, 22) have not run: stages 2-4 are not frozen or executed. Stage 1 of 4 is
complete, and the search archives are only partly complete (see the table). So no per-cell ADVANCES /
MATCHES / REGRESSES / TRADES can be assigned, and the rule's own "otherwise" branch applies.
Nothing here is a speedup claim, and nothing is credited to the search.

Distance to matched rho: **unchanged and not measured by this item.** It is a one-stage diagnostic
(S_4 pair-and-root kernel, toy factor bases, l = 9, n = 31, 37, 41); it computes no S, no rho ratio and
no GAE, so the section 5a pricing band (0, 0.1, 1 GAE) has no cross-method row to rank. The unpriced native
counters (word XORs, hash probes, trace tests, pair iterations) are priced at 0, 0.1 and 1 counted
operations inside Q for every arm, the same way. Even a full success would be a constant factor on
the relation phase, not the exponent.

First run proper (before the R2 pass)

The first run left no assessment: no README existed. It produced receipts for card 01 (selftest, completed) and
card 04 (n31 arm b, evaluations 0..191, completed), both kept and not re-run. Cards 02 and 03 were cut off
without a receipt; a partial `raw/state/n31-p.jsonl` (151 evaluations) has no receipt and is **not used**
(its evaluations 0..127 agree field for field with the R2 card-02 archive, which is consistent with the
evaluations being pure functions of the index; noted, not relied on). Cards 05-18 never ran. No first-run
verdict exists to supersede;
the status was "incomplete, nothing assessable".

What ran (R2 pass, all relays checked against receipts)

card	content	status	archive (sha256 prefix)
01 (first run)	build + selftest	completed	`raw/selftest.txt`
04 (first run)	n31 arm b, evals 0..191	completed	`raw/state/n31-b.jsonl` 23792d0b
02-r2	n31 arm p, evals 0..127	completed	`raw-r2/state-1/n31-p.jsonl` 83fac4f4
05-r2	n31 arm b, evals 192..255 (copy of card 04 archive, bound by sha, then appended)	completed	`raw-r2/state/n31-b.jsonl` 8d4e27e9
06-r2	n37 arm b, evals 0..127	completed	`raw-r2/state-1/n37-b.jsonl` 0c8c0d54
07-r2	n41 arm b, evals 0..127	completed	`raw-r2/state-1/n41-b.jsonl` 59c6c7fc
08-r2	n31 arm c, evals 0..127	completed	`raw-r2/state-1/n31-c.jsonl` e44de45a

| 10-r2 | n37 arm c, evals 0..127 | completed | `raw-r2/state-1/n37-c.jsonl` 0154f724 |
| 12-r2 | n41 arm c, evals 0..127 | completed | `raw-r2/state-1/n41-c.jsonl` ebaf9b23 |

Relay check (`tools/task_card.py check-relay`, reports in `raw/relay-1..7.txt`): cards 02, 06, 07, 08, 10, 12-r2
`relay_verbatim: true`, no findings. **Card 05-r2: check-relay returned `CARD FAIL`, not a relay finding:** its `copy-card-04-archive` command names the destination file `raw-r2/state/n31-b.jsonl`, which the card itself
created, and the linter (run after the fact) now reports it as an unpinned repository file. Checked by hand instead: the relayed card sha, receipt sha (e8a6d9a4...) and output sha (8d4e27e9...) equal the on-disk values,
the bound input `raw/state/n31-b.jsonl` still hashes to 23792d0b..., and the first 192 lines of the new archive are
byte-identical to it. This is reported as a deviation of the check, not repaired; the data are usable.
Not run yet: cards 03, 19, 20, 09, 11, 13 (stage 2: the second 128 evaluations of p-n31 b-n37 b-n41 c-n31 c-n37 c-n41),
14-18 and 21 (stage 3), 22 (stage 4).

Numbers (training set only, Q at the stated weight; measured, from the archives)

Training = 6 natural + 32 planted targets per base. The incumbent is evaluation 0 of each arm-b archive (Q at w = 0.1: n31 4162.2, n37 3264.2, n41 3663.4; planted retention phi = 1). "ratio" is the best finite training Q of
the arm divided by the incumbent's Q on the same targets. These ratios use the targets the programs were selected on, have no interval, and are exactly the quantity the rule forbids reading as a verdict (selection among 128-256 candidates on 6 natural targets, so they are optimistically biased).

cell, arm	evals done / 256	best program (training)	Q at w=0.1	ratio at w = 0 / 0.1 / 1	phi	dominated, Q=inf
n31 b	256	affine-aligned-s2 lift1 stop0 root0 quartic=b1(1) sqt1 ab=2 bt=1 (idx 94)	314.3	0.0753 / 0.0755 / 0.0767	0.94	2, 10
n37 b	128	affine-aligned-s1 lift1 stop1 root0 q1 sqt1 ab4 bt1 (idx 79)	331.8	0.1014 / 0.1016 / 0.1031	0.31	3, 5
n41 b	128	affine-aligned-s2 lift1 stop1 root0 q1 sqt1 ab4 bt1 (idx 66)	296.4	0.0807 / 0.0809 / 0.0821	0.34	0, 0
n31 c	128	affine-rand-s1 lift1 stop0 root0 q1 sqt0 pm1 ab4 bt1 (idx 99)	502.9	0.1207 / 0.1208 / 0.1219	0.44	26, 39
n37 c	128	sub-low lift1 stop1 root1 q0 ab4 bt1 (idx 28)	655.4	0.1979 / 0.2008 / 0.2172	0.50	48, 55
n41 c	128	affine-aligned-s1 lift1 stop0 root1 q1 sqt1 pm1 ab4 bt2 (idx 56)	389.5	0.1058 / 0.1063 / 0.1092	0.53	39, 44
n31 p	128	affine-aligned-s2 (same program as n31 b idx 94, found at idx 64)	314.3	start 15505 -> 314.3	0.94	0, 0

Running-best training Q (w = 0.1, recorded): n31 b 4162 -> 343 (64 evals) -> 314 (128) -> 314 (256); n37 b 3264 -> 418 -> 332;
n41 b 3663 -> 297 -> 296. All three search arms reach their plateau by evaluation 128 or earlier.
Search versus random on equal evaluations (128 each, training, w = 0.1): best-b / best-c = 0.625 (n31, b at 256),
0.506 (n37), 0.761 (n41). Descriptive only: no interval, no held-out, and arm c is half complete.

Controls (state as of now)

* Instrument correctness (selftest, `raw/selftest.txt`): 400 random programs against brute force at n = 17, l = 6,
  bad = 0; quartic forms equal the resultant; S_4 vanishes on x(P1 +- P2 +- P3) at n = 17, 31, 41; `#E` and
  `2E = ker delta` at n = 13..41; handicap checks 1.996 (pairs=full, 1.95-2.05), 0.213 (lift, 0.20-0.30), 0.234
  (abort 1/8, 0.22-0.25), 1.542 (generic root > 1.3), 0.9858 (b1 < 1): **all PASS**.
* Positive control for the search (arm p), provisional at 128 of 256 evaluations: best training Q 314.3 is at most the
  incumbent's 4162.2 and at most half of its start 15505. The pass criterion is met so far (ratio 0.0203 to its start),

  but the arm is not finished and the criterion is on the final 256, so it is recorded as "met provisionally".
* Negative control, arm c: half complete (128 of 256). Its best training Q is also far below the incumbent's
  (ratios 0.12, 0.20, 0.11), so random search over this program space already finds large training gains;
whether
  the search beats it is exactly the `Q(B)/Q(C)` comparison that needs held-out data.
  `affine-anti` never receives a finite Q wherever it was drawn (arms b n31, b n37 and the c archives):
consistent
  with its prediction.
* Null (incumbent on two held-out halves), yield conservation at (17,6) and (19,7), finalists, ablation: **not
run**.
  A finalist whose family violates the yield model cannot support a verdict, and that check has not happened.

Observations that the final held-out stage must test (not findings)

1. The best training programs share `lift = 1` (the incumbent's `lift = 0` scans abscissae without a point):
   the best `sub-low` program in the n31 b archive (lift1, otherwise incumbent-like) already has Q 834 (ratio
0.20).
   The reference arm a1 (incumbent plus `lift = 1, stop = 1`) is a card-14 output and has not run, so how much
of the
   gain is "do not scan non-rational abscissae" is **not yet separated**; the rule reports `Q(B)/Q(A1)` for
this.
2. The winners are `affine-aligned` bases. In Q this earns `T_eff` credit (n31: 2.90e6 versus 1.11e6; n37:
3.21e6
   versus 1.60e6; n41: 3.10e6 versus 1.35e6, factors 2.6, 2.0, 2.3), which is the yield model, not a cheaper
kernel;
   whether the model holds at those bases is the conservation check, which has not run. Part of the training
gain
   is therefore contingent on an unrun check.
3. The aligned bases give up planted retention (phi 0.94, 0.31, 0.34 for the three winners): Q divides by phi.
The
   held-out phi is what counts.

Deviations and caveats

* Staging, not a protocol change: the R2 pass put all outputs under `raw-r2/` and split searches into
128-evaluation
  pieces across cards. The addendum to `PROTOCOL.md` records this; question, arms, cells, metric and rule
unchanged.
* The relay check of card 05-r2 fails only the after-the-fact lint (above).
* The partial first-run `raw/state/n31-p.jsonl` (151 evaluations, no receipt) is not used.
* Host L0 (shared 4-vCPU container, two lanes at once): wall times in the archives are descriptive only; the
  verdict quantity is counted operations. Toy scale: GF(2^n), n = 31..41, l = 9, sparse regime (about 0.02 raw
  S_4 roots per natural target); a replica of the incumbent kernel, not the crate's `SubspaceOracle`
(Deviation 1);
  no result here transfers to cryptographic sizes.
* Not a performance result in the sense of AGENTS.md: no held-out measurement exists, so nothing was copied to
  `benchmarks/` and no leaderboard section was regenerated. Re-evaluate after stage 3.

Next steps (not done here)

Commit the stage-1 outputs (this commit), run `make_r2_cards.py 2`, execute stage-2 cards, then `3` and `4`;
then apply the rule to the held-out summary (`raw-r2/summary.json`, card 22) and replace this verdict.

Files

`graph.svg`, `graph.png` (source `make_graph.py`, data `raw-r2/training-stage-summary.json`), `REPORT.pdf`,
`raw/relay-1..7.txt`, `raw-r2/receipt-card-*-r2.json`, `raw-r2/state*/`.